

Assignment 2 | Sample Output #2

Quantum Computing and its Future Applications

A Research Report

Student Name	System ID	Date	Tools
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SECTION 01

Introduction

Classical computers store and process information as bits, each of which is either a 0 or a 1. Quantum computers use a different physical approach. Their basic unit, the qubit, can represent 0 and 1 at the same time through a property called superposition. Two or more qubits can also be linked through entanglement, so that the state of one instantly relates to the state of another. These properties allow quantum computers to evaluate many possible solutions at once for certain kinds of problems.

The technology has been under development for several decades, but practical progress has accelerated in the past few years. In 2019 a research team demonstrated that a quantum processor could complete a specific calculation in about three minutes that would have taken a conventional supercomputer thousands of years. Since then, processor sizes have grown and error rates have dropped, though significant engineering problems remain.

This report looks at what quantum computing can currently do, what the most credible near-term applications are, what stands in the way of wider use, and how the field is expected to develop over the coming decade. The research was collected through automated web search and Wikipedia queries.

SECTION 02

Agent Execution Log

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AGENT LOG -- Topic: Quantum Computing and its Future Applications

[THOUGHT] Need to cover fundamentals, current hardware state, industry applications, challenges.

[ACTION] web_search --> 'quantum computing 2025 progress applications industry'

[OBS] Google Willow chip sets new benchmarks. IBM roadmap targets 100k qubits by 2026. Market at $450B by 2035...

[ACTION] wikipedia_search --> 'Quantum computing'

[OBS] Superposition, entanglement, interference. Algorithms: Shor, Grover. Hardware types: superconducting, trapped ion...

[ACTION] web_search --> 'quantum computing challenges error correction 2025'

[OBS] Decoherence main barrier. NIST post-quantum standards finalized 2024. Error correction needs 1000:1 qubit ratio...

[THOUGHT] Enough data collected. Passing to report generation step.

[FINAL] Research complete.
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SECTION 03

Key Statistics

\$450B	1,000+	10^25x	2030
Projected market size by 2035	Physical qubits on IBM systems (2025)	Speed advantage claimed by Google Willow	Estimated fault-tolerant timeline

SECTION 04

Key Findings

1. Benchmark Results Keep Improving Google's Willow chip completed a test problem far faster than any classical machine while also showing that adding more qubits reduced errors rather than increasing them, which had been a longstanding concern.	2. Drug and Materials Research Quantum processors can model how atoms and molecules interact at a level of detail that classical computers struggle to reach. Pharmaceutical companies are using early quantum tools to screen drug candidates and study protein behavior.
3. Encryption Standards Updated The U.S. standards body finalized its first post-quantum encryption standards in 2024. The move was driven by concern that a sufficiently powerful quantum computer could break the encryption methods most of the internet currently relies on.	4. Financial Optimization Some financial firms have tested quantum and quantum-inspired algorithms for portfolio balancing problems. Early results show improved outcomes for problems with many competing constraints, though most work remains in research settings.
5. New Material Discovery A major cloud provider published results in 2025 showing that quantum simulation identified several candidate battery materials that had not been found through classical modeling approaches.	6. Quantum Communication Tests Researchers in the Netherlands demonstrated a quantum network linking multiple nodes over 50 kilometers of fiber, distributing entanglement in a way that makes communication theoretically immune to interception.

SECTION 05

Challenges

01

Decoherence

Qubits lose their quantum state very quickly when they interact with heat or electromagnetic noise. Most superconducting quantum computers must operate at temperatures far colder than outer space. Keeping qubits stable long enough to complete useful calculations is the central hardware problem.

- 02

Error Rates

Current processors make errors at rates that are too high for most practical tasks. Correcting these errors requires adding many extra physical qubits for every logical qubit, which means useful fault-tolerant machines will need millions of physical qubits.
- 03

Shortage of Skilled People

Building and programming quantum computers requires knowledge across physics, engineering, and computer science. The number of people with this combination of skills is small relative to demand, and training programs are still maturing.
- 04

Limited Useful Algorithms

There are only a small number of quantum algorithms that have been shown to offer a genuine advantage over classical methods. For most everyday computing tasks, classical hardware remains faster and more practical.
- 05

Cost and Access

Building quantum hardware is expensive and technically demanding. Cloud-based access has opened the technology to more researchers, but running large-scale experiments still requires resources that most organizations do not have.

SECTION 06

Future Scope

2026 - 2027	Fault-Tolerant Processors Several major hardware programs are targeting processors that can correct their own errors reliably. If achieved, this would open up chemistry and materials simulation at commercially relevant scales.
2027 - 2029	Hybrid Computing Pipelines Most near-term applications will combine classical and quantum processing. Cloud platforms are expected to offer quantum compute as a standard resource that software can call alongside conventional hardware.
2029 - 2032	Machine Learning Applications Researchers are studying whether quantum methods can speed up certain machine learning calculations. Early theoretical results suggest potential for specific linear algebra operations used in model training.
2030 and beyond	Secure Communication Networks Several governments are funding national quantum networks based on quantum key distribution. These systems would make it physically detectable if someone tried to intercept a communication.

SECTION 07

Conclusion

"Quantum computing will not replace classical hardware. It will handle specific problems that classical computers cannot solve within reasonable time or cost."

Quantum computing has moved from theoretical work to a field with working hardware, research results, and early commercial pilots. The benchmark achievements of recent years show that the underlying physics works. The remaining challenges are primarily about engineering at scale and finding problems where quantum advantage is large enough to justify the cost.

The most credible near-term applications are in chemistry, materials science, and encryption. These are areas where the problems are well defined and the advantage of quantum simulation is understood theoretically. Progress in these areas will likely drive the first commercially useful results.

Organizations with a long-term interest in computation-heavy research, financial modeling, or data security should be tracking this field and starting to build relevant knowledge. The technology will not be broadly available tomorrow, but the timeline for meaningful capability is measured in years, not decades.

SECTION 08

References

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